

Graphs of exact colorings

OPG-57824

L1 - TRACTABLE

PARTIALLY SOLVED

Algebra

ORIGINAL PROBLEM STATEMENT

Conjecture For $c \geq m \geq 1$, let $P(c, m)$ be the statement that given any exact c -coloring of the edges of a complete countably infinite graph (that is, a coloring with c colors all of which must be used at least once), there exists an exactly m -colored countably infinite complete subgraph. Then $P(c, m)$ is true if and only if $m = 1$, $m = 2$, or $c = m$.